

Proactive Price Beat System

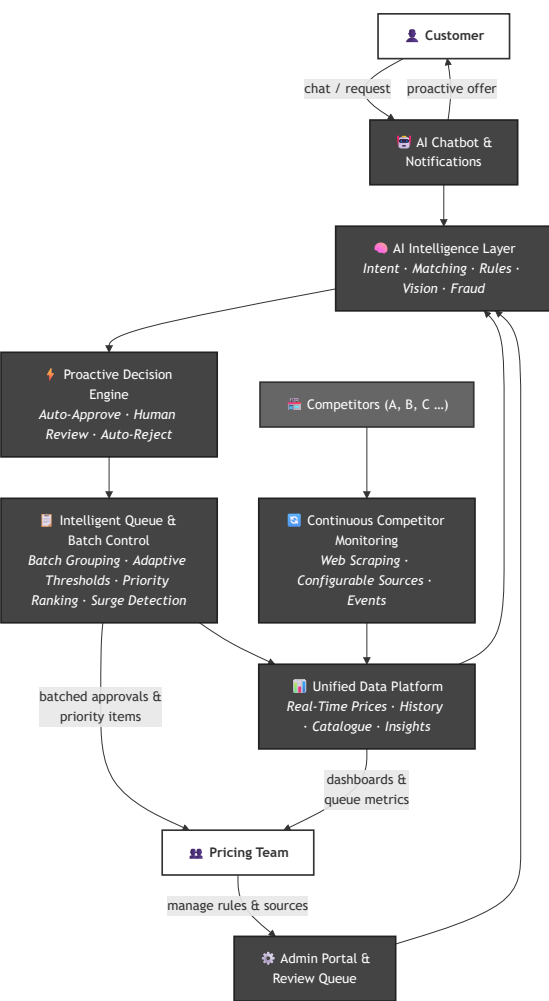
Solution Architecture — AI-First with Unified Data Platform

Azure AI Foundry · Semantic Kernel · Microsoft Fabric · Azure Bot Service

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1. Conceptual Architecture

High-level view showing how the system works — technology-agnostic, designed for executive audiences.



How It Works

1. **Competitors are continuously monitored** — The system scrapes competitor prices on a configurable schedule from an extensible URL registry
2. **All pricing data is unified** — Competitor prices, our prices, and product catalogues live in one platform with full versioned history
3. **AI evaluates every price change** — The intelligence layer automatically matches products, detects promotions, validates conditions, and determines if we should beat the price
4. **Risk-based decisions** — Low-risk beats are auto-approved instantly; complex or unusual cases are routed to the pricing team for human review
5. **Intelligent queue handles volume at scale** — Batches related decisions together, prioritises by revenue impact, and adapts thresholds during surge events so the team only reviews what truly needs human judgement
6. **Customers are proactively notified** — Before they even ask, we offer them the better price via email, SMS, or in-app notification
7. **Customers can also ask** — The AI chatbot handles reactive requests in real-time (URL, screenshot, or coupon code submission)

Component Descriptions

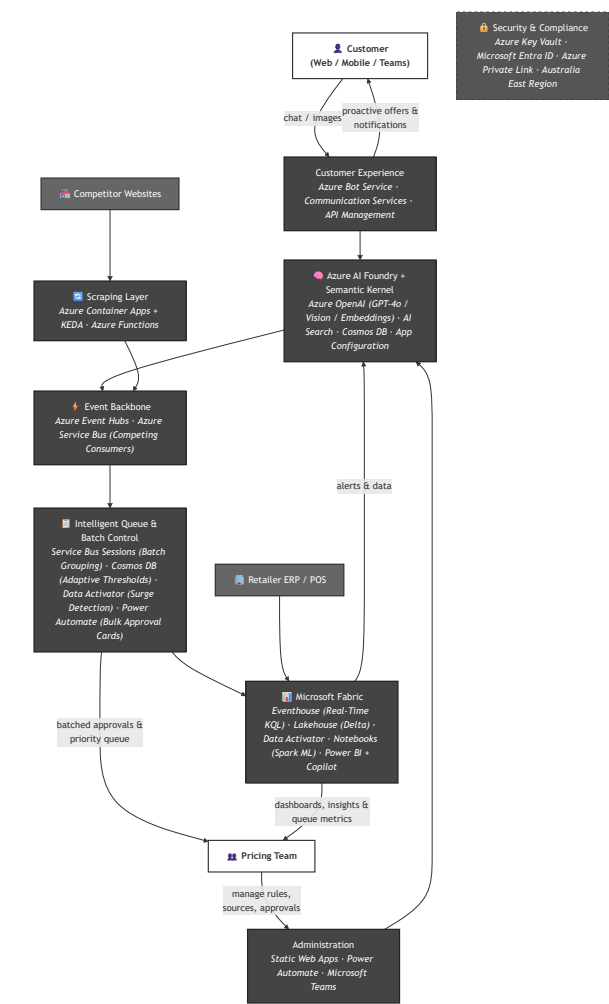
Component	Purpose
 AI Chatbot & Notifications	Customer-facing conversational AI. Handles both proactive notifications (system finds a beat) and reactive requests (customer submits a competitor price). Supports text, images, and URLs.
 Admin Portal & Review Queue	Internal tool for the pricing team to manage competitor sources, configure LLPG rules, toggle features, and review/approve medium-risk price beat decisions.
 AI Intelligence Layer	Core AI engine performing: customer intent detection, product matching (brand/variant/size), LLPG rule evaluation (20+ business rules), image/label analysis for wine vintages, and fraud detection.
 Proactive Decision Engine	Classifies each price beat opportunity by risk level. Low-risk = auto-approve and notify customer. Medium-risk = route to human review queue. High-risk (fraud/excluded) = auto-reject and log.
 Intelligent Queue & Batch Control	Handles volume at scale. Groups related decisions into batches for bulk approval, dynamically adjusts auto-approval thresholds during surges, prioritises review items by revenue impact, and routes categories to specialist reviewers.
 Unified Data Platform	Single source of truth combining real-time competitor prices (versioned with timestamps), historical price trends, unified product catalogue, and market insight analytics.

 Continuous Competitor
Monitoring

Always-on scraping pipeline. Competitor URLs are configurable and extensible (add new competitors without code changes). Events flow through a real-time pipeline into the data platform.

2. Azure Solution Architecture

Detailed view with specific Azure and Microsoft services mapped to each layer of the solution.



Azure Services by Layer

Layer	Azure / Microsoft Services	Purpose
Customer Experience	Azure Bot Service, Azure Communication Services, Azure API Management	AI chatbot (Web/Mobile/Teams channels), email/SMS notifications, secure API gateway with OAuth
Administration	Azure Static Web Apps, Power Automate, Microsoft Teams	Admin portal for URL/rule management, approval workflows for price beat review, Teams-based notification queue
AI Intelligence	Azure AI Foundry, Semantic Kernel, Azure OpenAI (GPT-4o / Vision / Embeddings), Azure AI Search	Multi-agent orchestration, LLM-powered intent/matching/rules, image analysis for screenshots & labels, vector search for product matching & LLPG knowledge base
Configuration	Azure Cosmos DB, Azure App Configuration	LLPG rule parameters and competitor source registry (Cosmos DB), feature flags and rule toggles (App Configuration)
Event Backbone	Azure Event Hubs, Azure Service Bus (Competing Consumers)	High-throughput price event streaming (Event Hubs), ordered transactional messaging for CHUB escalation (Service Bus), competing consumers for parallel team processing
Queue & Batch Control	Service Bus Sessions, Cosmos DB (thresholds), Data Activator (surge), Power Automate (bulk cards)	Batch grouping via sessions, adaptive auto-approval thresholds, real-time surge detection, Teams bulk approval cards for batch decisions
Unified Data Platform	Microsoft Fabric — Eventhouse, Lakehouse, Data Activator, Notebooks, Power BI + Copilot	Real-time KQL queries on price streams, Delta table historical archive, automated alert reflexes on price drops, Spark ML for trend analysis, natural language dashboards
Scraping Layer	Azure Container Apps + KEDA, Azure Functions	Auto-scaling Playwright scraping workers (scale to zero, burst at peak), timer-triggered scheduling and data quality normalization
Security & Compliance	Azure Key Vault, Microsoft Entra ID, Azure Private Link, Australia East Region	Secrets management, identity & access control, private networking for all PaaS services, Australian data residency compliance

Key Architecture Principles

- **Event-Driven:** Every state change (scrape result, price drop, rule decision, escalation) flows as an event through Event Hubs — no polling, no synchronous coupling
- **AI-First:** Semantic Kernel orchestrates all intelligent decisions — intent detection, product matching, rule evaluation, fraud scoring, and trend analysis

- **Unified Data:** Microsoft Fabric serves as the single data platform — real-time (Eventhouse) and historical (Lakehouse) in one place, with built-in ML and BI
- **Proactive + Reactive:** System continuously finds price beat opportunities (proactive) while still supporting customer-initiated requests (reactive)
- **Human-in-the-Loop:** Risk-based routing ensures complex decisions get human review via Power Automate / Teams approvals before reaching customers
- **Configurable & Extensible:** New competitor URLs, rule changes, and feature toggles require zero code changes — managed through admin portal and App Configuration
- **Scale-Ready:** Intelligent queue batches related decisions, adapts thresholds during surges, and routes by category — so the pricing team handles thousands of events without being overwhelmed

3. LLPG Rules Engine — Business Rules

The Lowest Liquor Price Guarantee (LLPG) Rule Engine evaluates 15+ business rules to determine if a competitor price qualifies for a price beat. Each rule is implemented as a Semantic Kernel plugin — independently testable, configurable without code changes.

Rule	What It Does
Beat Price Formula	Calculates the price to beat the competitor: 50c below for items under a threshold, \$1 below for items above (configurable)
Normalised Unit Price	Converts competitor prices to comparable unit price (per-litre, per-can). Translates cart-level discounts to product level
Retailer Allow-List	Rejects the claim if the competitor is excluded (e.g., duty-free, wholesale-only)
Stock-for-Stock	Caps beat quantity to match competitor's available stock — checked across any store nationally
Promo Limit	If competitor promo only applies to limited quantity (e.g., max 2), beat price applies to same quantity
Delivery Fee Exclusion	Excludes delivery fees from comparison. Exception: "spend X, get free delivery" promos
Vintage & Label Matching	Strict matching for wine vintages and visual labels (baseline vs. premium) using AI Vision
Lightning Sale Validation	Validates that flash sales / happy hours are actually active at the exact time of processing
Member Discount Rejection	If competitor price is members-only (loyalty/club), the price beat does not apply
Clearance Detection	Identifies clearance items at competitors and responds with relevant messaging
Radius Rule	10km radius rule (currently suspended, toggleable on/off via feature flag without code change)
RSA Compliance	Enforces Responsible Service of Alcohol floor price — alcohol cannot be sold below legal minimum
Fraud Risk Scoring	ML-based risk scoring to detect abuse (repeated claims, manipulated screenshots, fake URLs)

4. Scaling Strategy — Handling High-Volume Price Beat Events

When a competitor runs a site-wide sale or multiple competitors change prices simultaneously, thousands of beat decisions can arrive within minutes. This section explains how the architecture handles that at scale.

The Scale Challenge

Scenario	Estimated Volume
Single competitor site-wide sale (e.g., 20% off all spirits)	~500–2,000 decisions simultaneously
Multiple competitors change prices on the same day	~3,000–10,000 decisions
Holiday season (Boxing Day, EOFY)	Sustained high volume over days
New competitor added to monitoring	Initial backfill of entire catalogue

5-Layer Scaling Strategy

#	Strategy	What It Does	Azure Implementation
1	Batch Grouping	Groups related decisions by trigger (same competitor + same event). Team reviews the batch policy once → all items inherit the decision.	Service Bus sessions ; Cosmos DB batch metadata
2	Adaptive Auto-Approval	Dynamically adjusts confidence thresholds. Normal: >95% auto-approve. Surge (queue >500): lowers to >85% + post-audit flag.	Cosmos DB threshold config; Azure Functions recalculates
3	Priority Queue with SLA	Scores items by revenue impact × time sensitivity. High-value surfaces first; low-value auto-approves after timeout.	Service Bus priority queues ; Cosmos DB scoring
4	Category-Based Routing	Wine/vintage → wine expert. Spirits/beer → general team. Fraud → fraud analyst.	Service Bus topic subscriptions with category filters

5	Surge Detection	Monitors queue depth & arrival rate in real-time. Triggers team alerts, mode switches, staffing signals.	Fabric Data Activator ; Power BI real-time dashboard
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Batch Approval Example

1. Competitor BWS drops prices on 800 spirit products by 15%
 2. Scraping layer detects 800 price changes → 800 events through Event Hubs
 3. AI evaluates all 800 → produces 800 beat decisions
 4. **Batch Grouping** recognises all 800 share the same trigger and creates **1 batch**
 5. Pricing team receives **1 approval card** in Teams: *"BWS spirits sale — 800 products, avg beat \$2.40, total margin impact \$1,920. Approve all?"*
 6. Team clicks **Approve All** → 800 customers notified instantly
 7. Edge cases (vintage wine, clearance) are separated as individual items for dedicated review

Adaptive Mode Switching

Queue Depth	Mode	Behaviour
< 100	Normal	Strict thresholds, all medium-risk → human review
100 – 500	Elevated	Relax thresholds slightly, batch similar items
> 500	Surge	Aggressive auto-approve, post-audit, bulk cards, team alerts

All mode transitions are logged and auditable. Post-audit reports are generated automatically for decisions auto-approved during surge mode.